

Segmenting a image into meaningful regions by training a CNN with self-supervised losses

1 Problem Definition

This project aims to segment a single unlabeled image into meaningful regions using a self-supervised learning approach guided by edge information. The input is an RGB image $I \in \mathbb{R}^{H \times W \times 3}$, and the output is a segmentation mask $M \in \{1, \dots, K\}^{H \times W}$, where each pixel is assigned to a visual group. To enhance structural details, Sobel edge filters are applied to highlight important edges before extracting small patches $\{P_i\}_{i=1}^N$. A convolutional neural network is then trained using contrastive loss to learn feature representations from these patches. Finally, K-means clustering groups similar patches into coherent segments. This method is valuable because it eliminates the need for labeled data while still capturing meaningful visual structures. Success is assessed both quantitatively using intra-cluster variance and qualitatively through perceptual coherence of the output masks.

2 Related Work

Caron et al. [1] introduced *DeepCluster*, a pioneering method that combines self-supervised feature learning with clustering, where CNN features are iteratively grouped using K-means, and the cluster assignments serve as pseudo-labels for further training. While effective for large datasets, DeepCluster requires multiple images and complex re-labeling steps. In contrast, our approach operates on a single image, simplifying the pipeline and focusing on intra-image semantic consistency. Similarly, Van Gansbeke et al. [2] proposed *Unsupervised Semantic Segmentation by Contrasting Object Mask Proposals*, which leverages object-level region proposals and a contrastive loss for unsupervised segmentation. However, their method assumes access to pre-trained models and proposal generators, which introduces external dependencies. Our method differs by employing a simple CNN trained from scratch using contrastive loss solely on image patches, without any reliance on pretrained backbones or external region proposals, making it more applicable in fully unlabeled and data-scarce scenarios.

3 Method

We propose a four-stage, self-supervised pipeline that converts a *single* RGB image $I \in \mathbb{R}^{H \times W \times 3}$ into a segmentation mask without any labels. An overview is shown in Figure 1.

Edge-Aware Preprocessing: To highlight important structural features in the image, we apply Sobel filters to the grayscale version of the input image. Given horizontal and vertical gradients S_x and S_y , we compute the edge magnitude: $S = \sqrt{S_x^2 + S_y^2}$. This edge map is then converted to a 3-channel image and optionally enhanced using techniques such as CLAHE (Contrast Limited Adaptive Histogram Equalization) and image sharpening. We define four such variants—raw Sobel, CLAHE-enhanced, sharpened, and CLAHE+sharpened—and use each to produce different patch datasets for training.

Patch Sampling and Augmentation: From each edge-enhanced variant, we extract N random patches $\{P_i\}_{i=1}^N \in \mathbb{R}^{p \times p \times 3}$ to construct a dataset for feature learning. Each patch is passed through a double augmentation pipeline, producing two views for contrastive learning. These views form positive pairs, while the rest of the batch serves as negatives. This ensures that the network learns to recognize patch similarity regardless of augmentation.

Contrastive Representation Learning: A simple convolutional neural network is used to extract features from each patch. Let $z = f_\theta(P)$ represent the feature vector of patch P , where f_θ denotes the CNN. The model is trained using a contrastive loss that encourages features from the same patch (augmented differently) to be similar, while pushing away features from different

patches. The contrastive loss is defined as:

$$\mathcal{L}_i = -\log \frac{\exp(\text{sim}(z_i, z_{i'}^+)/\tau)}{\sum_{j \neq i} \exp(\text{sim}(z_i, z_j)/\tau)}$$

where $\text{sim}(a, b)$ is cosine similarity, τ is temperature parameter and $z_{i'}^+$ is the positive pair of z_i .

K-means Clustering and Segmentation: Once trained, the CNN is used to generate feature embeddings from overlapping patches across the edge-enhanced image. To incorporate spatial context, we append scaled 2D spatial coordinates to each feature vector. These combined feature representations are then clustered using K-means into K groups. Each pixel in the image is assigned the cluster label of its corresponding patch, resulting in the segmentation mask $M \in \{1, \dots, K\}^{H \times W}$. It is evaluated using intra-cluster variance: $\frac{1}{N} \sum_{i=1}^N \|z_i - c_{l_i}\|^2$ where z_i is a feature or color vector, c_{l_i} is the center of its cluster, and N is the total number of samples.

4 Results

We tested our method on the image of the *John Curtin School of Medical Research at ANU*, without using any labels. We applied the full pipeline to four Sobel-enhanced versions of the image: raw Sobel, CLAHE, sharpened, and CLAHE+sharpened. For each variant, we randomly sampled 2,000 patches, trained a CNN using contrastive loss, and performed K-means clustering with $K = 7$ to produce the segmentation. Quantitatively, the performance was evaluated using intra-cluster variance. As shown in Figure 4, V0 (raw Sobel) had the lowest variance, indicating compact and consistent clustering, while V3 (CLAHE+sharpened) had the highest variance, reflecting less compact groupings. These findings are consistent with the qualitative results in Figure 5, where V0 produced the clearest and most distinct boundaries, and V3 resulted in more noisy and less meaningful segmentation. This suggests that stronger features, captured through suitable edge preprocessing, lead to better segmentation and lower variance.

An ablation study can find out which parts of the method matter the most. When Sobel edge enhancement was removed and raw image patches were used instead, the segmentation appeared visually pleasing but lacked fine-grained details (Figure 2). In contrast, when the contrastive loss was replaced with a dummy loss, the CNN failed to learn meaningful features, resulting in poor clustering and segmentation quality (Figure 3). These results confirm that contrastive learning is essential for effective feature learning, while edge preprocessing plays a supportive.

5 Reflection

This project shows that self-supervised segmentation can be useful in situations where labeled data is hard to get, such as in medical imaging, environmental research, or historical image analysis. Because it works without manual labels, it could help smaller teams or non-experts use image analysis tools more easily. However, there are also risks. If the method is used in important areas like healthcare or security without careful checking, it might give incorrect results or miss important details. Since the system learns only from visual patterns, it might also reflect biases in the image, such as lighting or texture differences. It is important to use this method responsibly and make sure the results are interpreted carefully.

6 Conclusion

We found that contrastive learning is crucial for training the CNN to extract meaningful features for segmentation, even from a single unlabeled image. Although Sobel edge enhancement sometimes improved results, it was not always necessary and its effectiveness varied by image. If we had more time, we would test the method on more images, experiment with different CNN architectures, and further study how edge information affects segmentation performance.

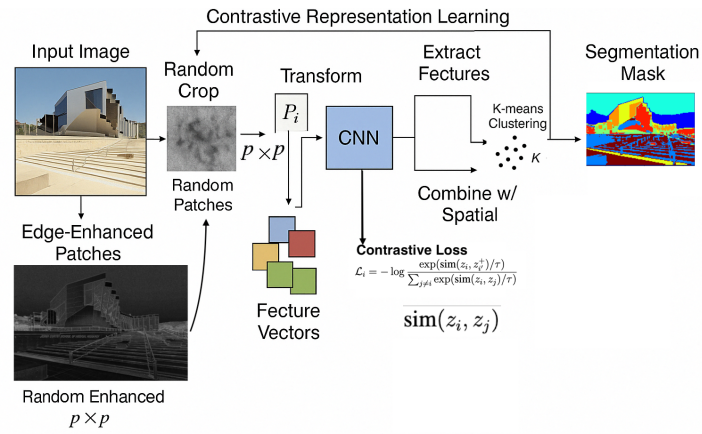


Figure 1: Overview of the proposed self-supervised segmentation method, including Sobel edge enhancement, patch sampling, CNN-based contrastive learning, and clustering-based segmentation.

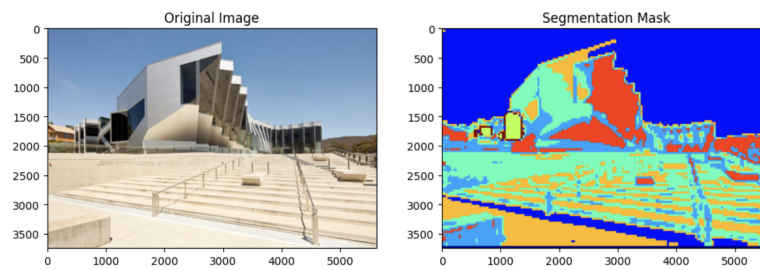


Figure 2: Original image VS the segmentation result using raw image patches without Sobel enhancement.

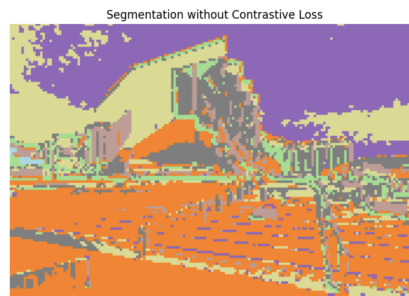


Figure 3: Segmentation result using a dummy loss instead of contrastive learning.

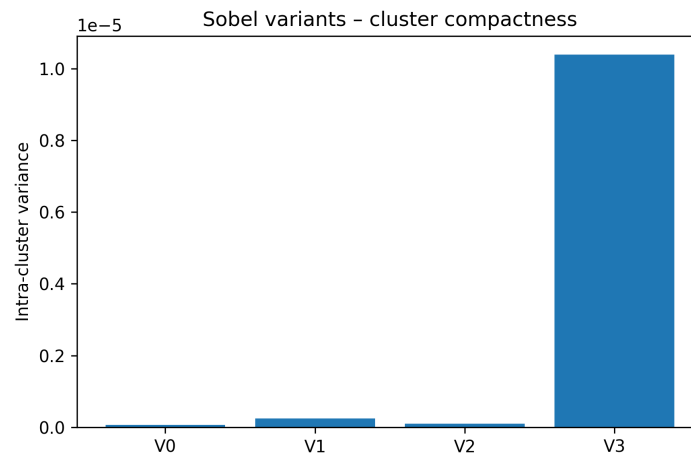


Figure 4: Intra-cluster variance of CNN feature representations across Sobel-enhanced variants V0–V3.

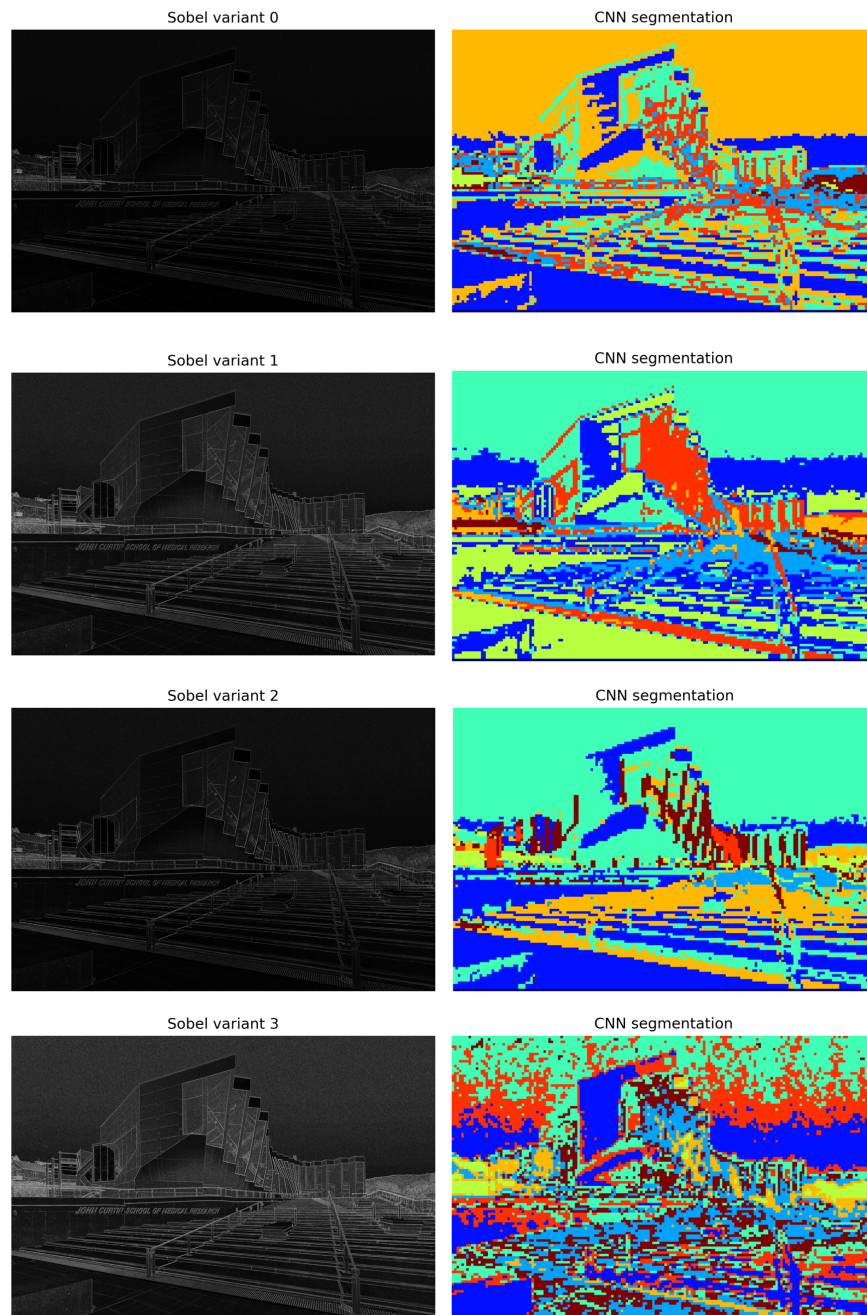


Figure 5: Visual comparison between Sobel-enhanced variants (left column) and their corresponding CNN-based segmentation results (right column). Each row represents a different Sobel variant: v0: raw Sobel, v1: CLAHE-enhanced Sobel, v2: sharpened Sobel, and v3: CLAHE followed by sharpening.

References

- [1] Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Matthijs Douze. Deep clustering for unsupervised learning of visual features. In *ECCV*, 2018.
- [2] Wouter Van Gansbeke, Simon Vandenhende, Stamatios Georgoulis, and Luc Van Gool. Unsupervised semantic segmentation by contrasting object mask proposals. In *ICCV*, 2021.